

# Algoding Research Comparison

This report compares the strongest rule-based, ML, and DL strategy branches under the execution-aware research path. The latest DL rerun disables the no-same-day reentry rule and extends the window set to 1m, 3m, 6m, 9m, 1y, 2y, and 3y.

## Headline Findings

- Best overall recorded strategy: SLV / slv\_torch\_cnn\_dl (DL, hourly cost-aware, no-same-day OFF), average total return 44.00%, drawdown -14.87%.
- Best DL strategy in the new no-rule rerun: SLV / slv\_torch\_cnn\_dl, average total return 44.00%, profitable windows 5/7.
- Across the matched DL set, disabling no-same-day reentry improved the average report score from 0.082 to 0.088.
- Shared execution-aware assumptions for the hourly branch: 5% fixed stop, 1.5% trailing stop, regular-hours-only bars, spread, market impact, stop slippage, regulatory fees, and partial-fill caps.

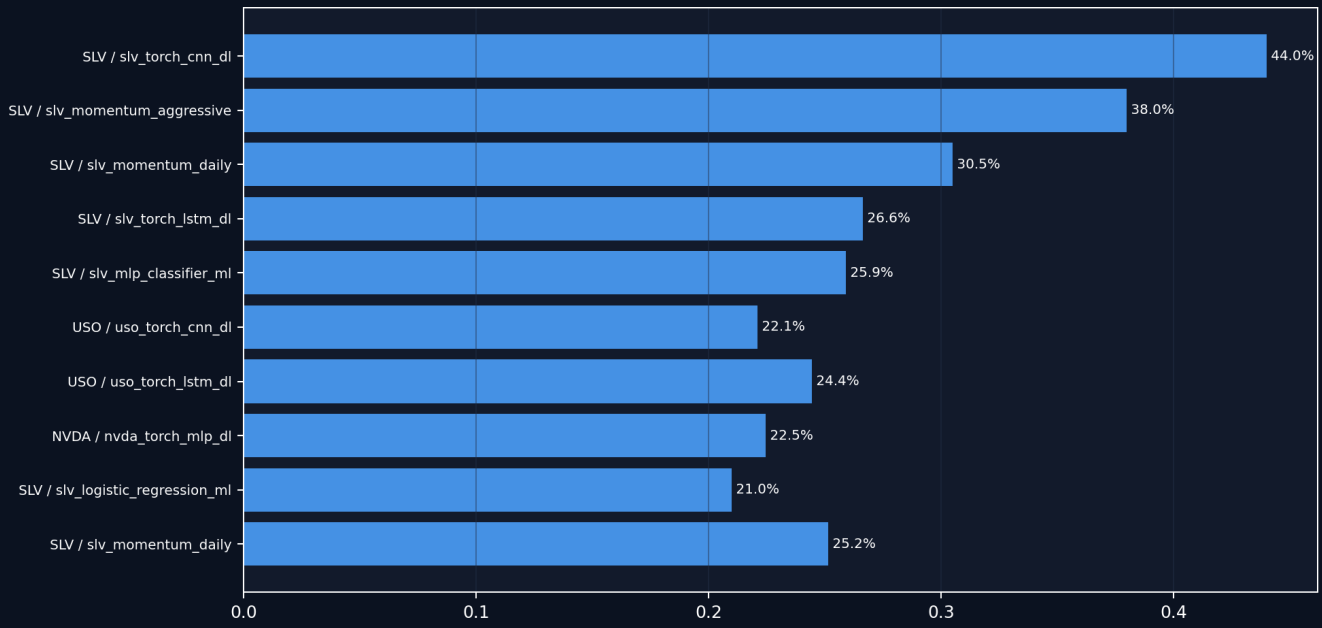
## Branch Leaders

| Branch                                          | Experiment                         | Symbol | Strategy                | Windows | Avg Return | Avg DD  | Score |
|-------------------------------------------------|------------------------------------|--------|-------------------------|---------|------------|---------|-------|
| DL / hourly cost-aware, no-same-day OFF         | hourly cost-aware, no-same-day OFF | SLV    | slv_torch_cnn_dl        | 7       | 44.00%     | -14.87% | 0.463 |
| DL / hourly cost-aware, no-same-day ON          | hourly cost-aware, no-same-day ON  | SLV    | slv_torch_lstm_dl       | 3       | 26.64%     | -10.98% | 0.341 |
| ML / hourly cost-aware, no-same-day ON          | hourly cost-aware, no-same-day ON  | SLV    | slv_mlp_classifier_ml   | 3       | 25.91%     | -10.31% | 0.335 |
| Rule-based / hourly cost-aware, no-same-day OFF | hourly cost-aware, no-same-day OFF | SLV    | slv_momentum_daily      | 6       | 25.16%     | -20.52% | 0.273 |
| Rule-based / hourly cost-aware, no-same-day ON  | hourly cost-aware, no-same-day ON  | SLV    | slv_momentum_aggressive | 6       | 37.98%     | -11.62% | 0.455 |

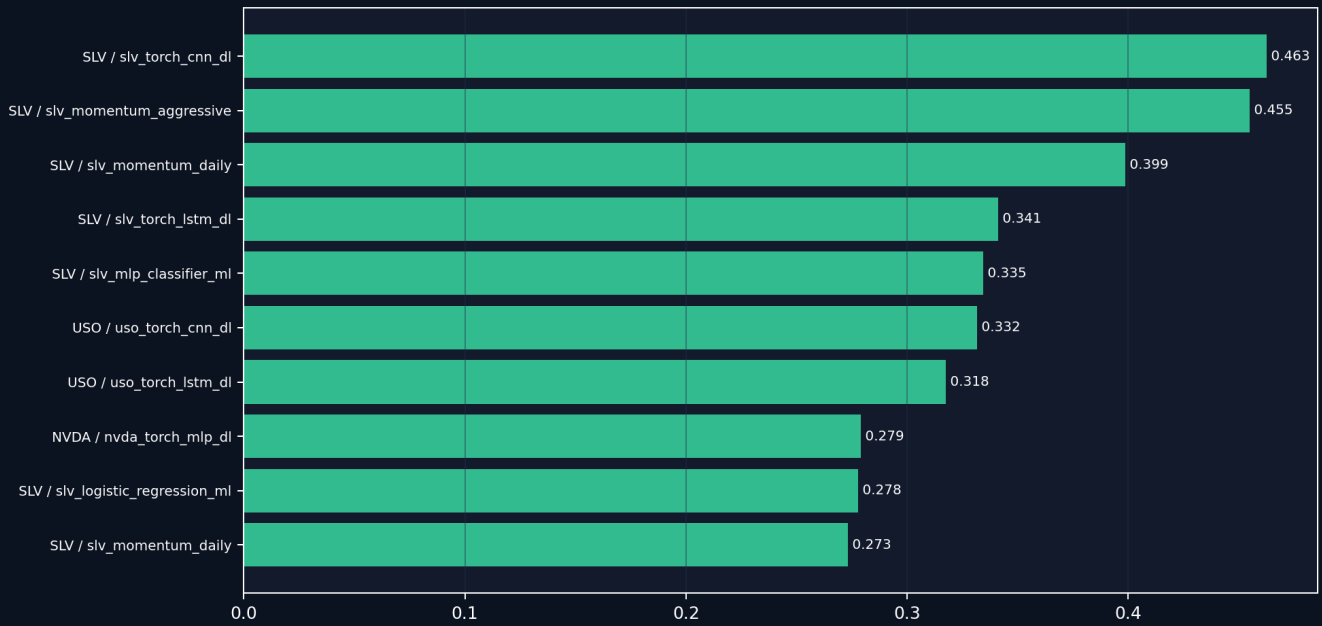
## Top Strategies Overall

| Rank | Branch         | Symbol | Strategy                   | Experiment                         | Win | Avg Return | Avg Ann. | Avg DD  | Score |
|------|----------------|--------|----------------------------|------------------------------------|-----|------------|----------|---------|-------|
| 1    | DL             | SLV    | slv_torch_cnn_dl           | hourly cost-aware, no-same-day OFF | 5/7 | 44.00%     | 41.83%   | -14.87% | 0.463 |
| 2    | Rule-based SLV | SLV    | slv_momentum_aggressive    | hourly cost-aware, no-same-day ON  | 6/6 | 37.98%     | 51.93%   | -11.62% | 0.455 |
| 3    | Rule-based SLV | SLV    | slv_momentum_daily         | hourly cost-aware, no-same-day ON  | 6/6 | 30.51%     | 50.99%   | -9.27%  | 0.399 |
| 4    | DL             | SLV    | slv_torch_lstm_dl          | hourly cost-aware, no-same-day ON  | 3/3 | 26.64%     | 19.49%   | -10.98% | 0.341 |
| 5    | ML             | SLV    | slv_mlp_classifier_ml      | hourly cost-aware, no-same-day ON  | 3/3 | 25.91%     | 15.24%   | -10.31% | 0.335 |
| 6    | DL             | USO    | uso_torch_cnn_dl           | hourly cost-aware, no-same-day OFF | 7/7 | 22.11%     | 97.23%   | -7.14%  | 0.332 |
| 7    | DL             | USO    | uso_torch_lstm_dl          | hourly cost-aware, no-same-day OFF | 7/7 | 24.43%     | 60.66%   | -9.18%  | 0.318 |
| 8    | DL             | NVDA   | nvda_torch_mlp_dl          | hourly cost-aware, no-same-day ON  | 2/3 | 22.46%     | 6.83%    | -7.23%  | 0.279 |
| 9    | ML             | SLV    | slv_logistic_regression_ml | hourly cost-aware, no-same-day ON  | 3/3 | 21.00%     | 14.19%   | -12.32% | 0.278 |
| 10   | Rule-based SLV | SLV    | slv_momentum_daily         | hourly cost-aware, no-same-day OFF | 6/6 | 25.16%     | 27.12%   | -20.52% | 0.273 |
| 11   | Rule-based USO | USO    | uso_momentum_daily         | hourly cost-aware, no-same-day OFF | 7/6 | 17.28%     | 35.22%   | -7.83%  | 0.270 |
| 12   | ML             | NVDA   | nvda_mlp_classifier_ml     | hourly cost-aware, no-same-day ON  | 2/3 | 22.52%     | 8.88%    | -10.39% | 0.264 |

Top Strategies by Average Total Return



Top Strategies by Unified Report Score

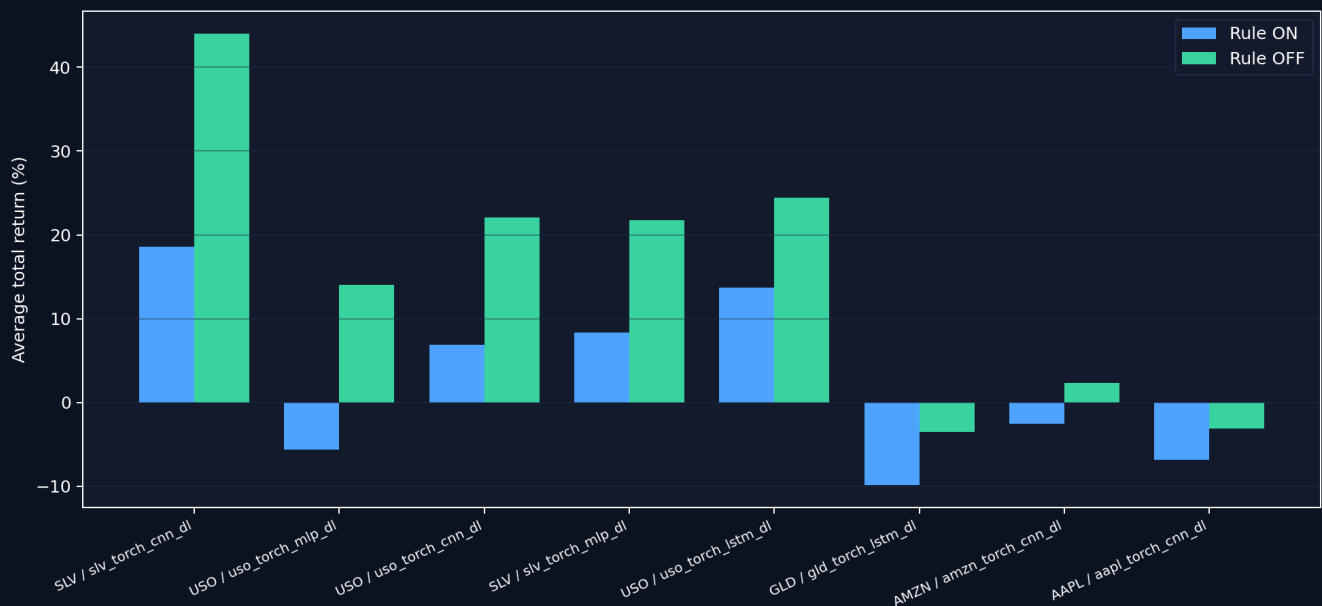


## DL Rule Toggle

This table compares the same DL strategies before and after disabling no-same-day reentry. Positive deltas mean the no-rule version improved the average total return.

| Symbol | Strategy           | With Rule | Without Rule | Delta   | With DD | Without DD |
|--------|--------------------|-----------|--------------|---------|---------|------------|
| SLV    | slv_torch_cnn_dl   | 18.56%    | 44.00%       | +25.43% | -12.82% | -14.87%    |
| USO    | uso_torch_mlp_dl   | -5.65%    | 14.05%       | +19.70% | -8.82%  | -5.64%     |
| USO    | uso_torch_cnn_dl   | 6.89%     | 22.11%       | +15.22% | -7.05%  | -7.14%     |
| SLV    | slv_torch_mlp_dl   | 8.39%     | 21.77%       | +13.38% | -8.86%  | -12.18%    |
| USO    | uso_torch_lstm_dl  | 13.74%    | 24.43%       | +10.69% | -7.13%  | -9.18%     |
| GLD    | gld_torch_lstm_dl  | -9.81%    | -3.54%       | +6.28%  | -15.64% | -12.43%    |
| AMZN   | amzn_torch_cnn_dl  | -2.52%    | 2.31%        | +4.83%  | -12.18% | -6.00%     |
| AAPL   | aapl_torch_cnn_dl  | -6.81%    | -3.09%       | +3.72%  | -12.08% | -7.07%     |
| MSFT   | msft_torch_cnn_dl  | -1.11%    | 1.17%        | +2.28%  | -5.77%  | -2.26%     |
| AAPL   | aapl_torch_mlp_dl  | -1.60%    | 0.12%        | +1.72%  | -8.63%  | -6.18%     |
| META   | meta_torch_lstm_dl | -5.16%    | -3.60%       | +1.56%  | -11.85% | -9.67%     |
| MSFT   | msft_torch_mlp_dl  | -3.17%    | -2.23%       | +0.94%  | -5.96%  | -5.66%     |

DL Average Total Return: No-Same-Day Reentry ON vs OFF



## Artifact Scope

- Rule-based comparison artifacts: reports\research\slv\_momentum\_daily\_hourly\_1p5\_no\_same\_day\_compare.json, reports\research\hourly\_no\_same\_day\_reentry\_contenders.json
- ML artifacts: reports\research\ml\_commodities\_hourly\_gpu\_run.json, reports\research\ml\_etfs\_stocks\_hourly\_gpu\_run.json
- DL with-rule artifact: reports\research\dl\_commodities\_stocks\_hourly\_gpu\_run.json
- DL no-rule artifact: reports\research\dl\_commodities\_stocks\_hourly\_gpu\_no\_reentry\_off\_7w\_run.json
- Unified report score formula used for cross-branch ranking:  $\text{avg\_total\_return} + 0.5 * \text{avg\_max\_drawdown} + 0.1 * \text{profitable\_ratio} + 0.02 * \min(\text{avg\_profit\_factor}, 3.0)$ .